Course notes

Revision: conditioning and independence

If $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. A branch of a tree carries a conditional probability; the probability of a path is the product of its branches. For a partition $A_1, \dots, A_r$, $P(B) = \sum_i P(A_i \cap B)$; if all the $P(A_i) > 0$, it is written $\sum_i P(A_i) P_{A_i}(B)$. Two events are independent when $P(A \cap B) = P(A)P(B)$. For $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. Do not confuse $P_A(B)$ and $P_B(A)$.

Example. $P(A) = 0.4$, $P_A(B) = 0.7$, $P_{\overline{A}}(B) = 0.3$. Then $P(B) = 0.4 \times 0.7 + 0.6 \times 0.3 = 0.46$ and $P_B(A) = \frac{0.28}{0.46} = \frac{14}{23} \approx 0.609$, which differs from $P_A(B) = 0.7$.

Bernoulli and the binomial distribution

A Bernoulli trial has two outcomes, success with probability p and failure with probability $1 - p$. The indicator variable B equals 1 in case of success and 0 otherwise: $E(B) = p$, $V(B) = p(1-p)$. A Bernoulli scheme is a repetition of n identical and independent trials. The number X of successes follows the binomial distribution $\mathcal{B}(n, p)$.

Key point · Distribution and parameters. For $k \in \{0, \dots, n\}$,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$E(X) = np, \qquad V(X) = np(1-p), \qquad \sigma(X) = \sqrt{np(1-p)}.$$

Proof of the distribution. Each path containing k successes has probability $p^k(1-p)^{n-k}$ by independence. There are $\binom{n}{k}$ choices for the positions of the successes. These paths are mutually exclusive; add their probabilities.

Cumulative events

$$P(X \leq k) = \sum_{j=0}^{k} P(X = j), \qquad P(X \geq k) = 1 - P(X \leq k-1).$$

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a-1).$$

For at least one success, $P(X \geq 1) = 1 - (1-p)^n$. The complement of "at least k" is "strictly fewer than k".

Example. For $X \sim \mathcal{B}(10, 0.2)$, $E(X) = 2$ and $V(X) = 1.6$. $P(X = 2) = 45(0.2)^2(0.8)^8 \simeq 0.3020$ and $P(X \geq 1) = 1 - 0.8^{10} \simeq 0.8926$. The expected value is neither a certain result nor necessarily the most likely one.

Threshold and interval

To find the smallest n such that $1 - (1 - p)^n \geq q$, a loop on n is enough. After taking logarithms, one can solve $(1 - p)^n \leq 1 - q$ then round in the right direction. To find the smallest integer k such that $P(X > k) \leq \alpha$, increase k until $P(X \leq k) \geq 1 - \alpha$ is obtained, then check the previous value. One can also look for an interval $[a, b]$ such that $P(a \leq X \leq b)$ is at least a prescribed value, without assuming that it is unique.

Independent trials, beyond two outcomes

For independent experiments, the probability of a list of results is the product of the probabilities of its components. The sample space is the Cartesian product of the sample spaces of each experiment. For example, for two independent throws of a fair die, the sample space contains the 36 pairs (i, j), $1 \leq i, j \leq 6$, each with probability $\frac{1}{36}$. The probability that the sum is 5 is $\frac{4}{36} = \frac{1}{9}$, because the favourable pairs are $(1, 4), (2, 3), (3, 2), (4, 1)$. In a Bernoulli scheme, success is coded by 1 and failure by 0: the sample space is $\{0, 1\}^n$. Independence concerns the whole succession; a mere equality between two observed probabilities is not enough to prove it.